

Early Detection of Heart Failure And Effective Optimized Feature Selection using Artificial Neural Network.

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I. *Abstract*

Globally, heart failure ranks as one of the top causes of mortality. This malady has a direct impact on the heart's performance. Nutrients and oxygen are not delivered and circulated effectively as a result of this disease. This paper is about developing a deep learning (DL) model which can predict heart failure conditions through Patient's checkup reports. Deep learning can yield greater precision compared to machine learning models because it is capable of managing large and complicated data. We use an Artificial Neural Network (ANN) due to its strong compatibility with Numerical and categorical data, which can be useful for extracting optimized features from patient data. By using Machine learning (ML) Algorithms, specifically, we use a decision tree algorithm, we can classify the outcome through Decision Trees, into multiple classes based upon the severity of the patient. This could help cardiologists to find heart failure in earlier stages and within a short period, which leads to saving patient's lives and money.

Keywords: Deep Learning, ANN, Machine Learning, Heart Failure

II. INTRODUCTION

The heart serves as a essential part within a human body tasked with circulating the blood through the entire circulatory system. It functions as a muscular pump that ensures oxygen-rich blood reaches all the organs and tissues while also removing carbon dioxide and waste products. Positioned slightly to the left of the chest, the heart serves a central role in sustaining human life.

Heart failure(HF) is a condition where the heart cannot effectively circulate the blood leading to heart problems[1][2]. Early diagnosis and Forecasting plays a vital position in enhancing patient outcomes and improve survival rates. Traditional methods for heart failure prediction often face challenges in handling complex medical data and capturing nonlinear relationships. However deep learning techniques have emerged as promising alternatives, offering enhanced predictive capabilities. Whereas Heart Failure (HF) should not be confused with a Heart Attack . HA happens when a blood clot obstructs one of the heart's vessels, impairing the heart's ability to pump blood effectively. In such situations, the heart may struggle to supply oxygenated blood to tissues, resulting in oxygen deprivation throughout the body. The severity of this circulatory failure typically depends on the extent of the heart attack. A heart attack (HA) is among the leading causes of HF. Key contributors to HF include coronary artery disease, inflammation of the heart, hypertension (high blood pressure), cardiomyopathy, or irregular heart rhythms. Additionally, other contributors such as sudden shocks, intense stress may precipitate either HF or heart attack. These events are inherently unpredictable.

Deep learning is a sub part of AI which is used for finding patterns in large datasets. The artificial neural networks (ANNs) are the most important part in the deep learning concept and it works as like a human brain. The ANNs consists of interconnected layers of neurons in the network each processing information and passing it on to subsequent layers. This hierarchical structure empowers ANNs to extract complex features including image recognition, natural language processing and in this case heart failure prediction. The main advantage of ANNs is their capability of capturing the non-linear association between the various factors providing a more accurate depiction of the underlying biological process involved in heart failure and also ANNs can handle large and various datasets, and obliging a wide range of clinical variables.

III. LITERATURE REVIEW

A Variety of approaches in machine learning and deep learning, including Artificial neural networks, Decision Tree and support vector machines have been widely studied in the literature to improve the accuracy of models.

C. B. Rjeily :- In this study[2] the author uses CPT+ algorithm for predicting the heart failure based

upon the Cleveland dataset, which consists of 207 rows. This dataset size may not be sufficient to predict heart failure in all cases worldwide and the CPT+ algorithm may take high computational cost when worked on larger datasets. The CPT+ reported an accuracy of 90.5%.

Shouman M :- This study [4] uses the Cleveland Heart Diseases Dataset with only 297 records, which may not be applicable to larger population. Finding the initial K value in K-Means is challenging, affecting the clustering performance. However, the author uses inlier, outlier and range methods for selecting the initial centroids. These methods may lead to mixed cases being grouped together, because this method finds the initial centroids based upon the smallest and largest values of attributes, increasing the risk of misclassification. However, the model accuracy obtained is 83.9%.

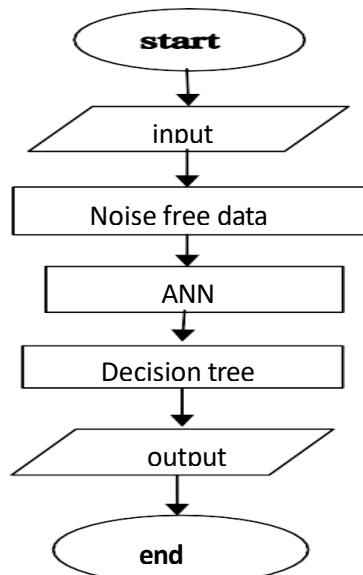
Priyanga :- This study uses the heart diseases dataset with only 303 records, limiting generalization. It relies on the naïve Bayes model, which assumes feature independence, a potentially unrealistic assumption for correlated medical data. The reported accuracy of 86 % is not compared with more advanced models [5].

Abdalla Mahgoub:- This study proposes two different comparative machine learning approaches to predict heart diseases cases in unknown data [1]. The reported average accuracy of Multilayer perceptron is 85.49%. More specifically, the Relu activation functions give better accuracy than other activation functions. According to the author's analysis, the Multi Layer perceptron yields better performance than other machine learning algorithms such as KNN, Logistic Regression and Naïve Bayes.

Marbaniang :- The study compares six machine learning for cardiovascular disease prediction, showing that KNN delivered the highest accuracy of 72.91%, succeeded by Random Forest at 72.12%. The drawbacks include computational time variations, with SVM being computationally expensive in terms of and naïve Bayes having lower accuracy of 60.1%. Adding BMI and Blood Pressure as features improved accuracy slightly [9].

H. Agrawal :- This study uses an ensemble model combining 10 Classifiers for heart failure prediction, achieving 85.2% test accuracy and 87.5% recall, while the ensemble improved recall, drawbacks include computational complexity and possible overfitting due to multiple models. Individual classifier performance varied, with logistic regression (86.4%), XGBoost (86.1%), and Random Forest (86.2%) showing high accuracy, whereas Naïve Bayes has lower accuracy (82.5%) [9].

IV. Proposed Algorithm



The proposed algorithms follow these steps to identify the heart failure outcome effectively. Initially, the input taken from UCI repository is then preprocessed to avoid noise data. The preprocessed data is passed to the ANN model, which processes the data and extracts the optimized features. In other words, it reduces the input data from 13 features to 8 features. These 8 optimized features then serve as input to the decision tree classifier to predict the outcome into two classes. The detailed methodology of the proposed system is discussed below.

V. METHODOLOGY

Here, we use an Artificial Neural Network (ANN) is employed as a proposed model to select important features within the data and then passed to Decision Tree classifies the heart failure into different classes derived from the severity levels of the patient. We will also follow several key steps to achieve an effective heart failure prediction system.

- a. Data collection.
- b. Data cleaning and preprocessing.
- c. Model selection.
- d. Feature selection.
- e. Model Evaluation
- f. Visualization.

We look step by step to achieve the heart disease prediction.

1. Data Collection

We utilized the most famous dataset which is Cleveland dataset of heart disease prediction provided by "Kaggle" website. Actually, this dataset comprises 76 attributes, yet all documented experiments utilize 14 attributes of them. Out of these 14 attributes, 13 features and the remaining one attribute tell us heart failure or not or we can say target variable. The shape of this dataset is (1025,14) it tells that 1025 different cases and 14 features. The entire data will be in integer and categorical values, the attributes are Age, trestbps, chol, thalach, Old peak, ca, thal, and targets hold integer values, while the remaining attributes are Sex, cp, fbs, restecg, exang, and slopes are categorical in nature.

Table 1:

Description of each feature mentioned here and these attributes plays a crucial role for predicting heart failure out of 76 attributes.

Attributes	Description
Age	Patients age in terms of years.
Sex	Patients Gender (male:1,female:0)
Cp	Patient's chest pain category: 0 for (typical angina), 1 for (atypical angina), 2 for (non- anginal pain), 3 for asymptomatic
Trestbps	The blood pressure level of the patient at rest (in mm/Hg)
Chol	cholesterol content in the blood in mg/dl
Fbs	Blood sugar levels during fasting above 120 mg/dl denoted as 1 for true and 0 for false
restecg	The resting ECG results are classified into three categories : 0 (Normal), 1(ST-T wave abnormality-T wave inversions or ST elevation/depression exceeding 0.05 mV), and 2 (probable/definite left ventricular hypertrophy by Estes criteria).
Thalach	peak heart rate achieved
Exang	Angina induced during exercise 0 indicates NO, 1 indicates Yes
old peak	ST-depression caused by exercise , measured against the resting state
Slope	During peak exercise, the ST segment slope is recorded as : 0- up sloping, 1 – flat, 2- down sloping
Ca	The total number of main vessels (0- 3)
Thal	A blood disorder thalassemia is indicated as : 0- NULL 1- normal blood flow 2- fixed defect (no blood flow in some part of the heart) 3- reversible defect (a blood flow is observed but it is not normal)
Target	target value: 0-no heart failure 1-heart failure.

The creators of this dataset listed below.

1. Hungarian Institute of Cardiology. Budapest: Andras Janosi, M.D. [8].

2. University Hospital, Zurich, Switzerland: William Steinbrunn, M.D. [8].
3. University Hospital, Basel, Switzerland: Matthias Pfisterer, M.D. [8].
4. V.A. Medical Center, Long Beach and Cleveland Clinic Foundation: Robert Detrano, M.D., Ph.D. [8].

2. Data Cleaning and preprocessing :-

The dataset used is the Cleveland Heart Disease dataset obtained from the UCI repository. Out of all attributes two features have missing values, those features are (ca, thal) to exclude the overfitting and underfitting condition of the ANN model the data needs to be noise-free. If not, the model failed to predict the heart disease of the patient correctly. All missing values were addressed by Python techniques, subsequently all missing values are eliminated. We will convert those missing values and NaN values into categorical values by using One-hot encoding, mean imputing, and KNN.

3. Model Selection

We have a lot of machine learning algorithms to classify heart disease, but they frequently failed to handle unstructured data. In contrast, supervised learning algorithms can categorize heart disease by using a target variable in the dataset, but this techniques gives best results when dealing with the labeled data. Deep learning algorithms gives a higher accuracy compared to typical learning algorithms and also handle the both structured and unstructured data. There is a wide array of deep learning models and we can select the best one by considering the problem statement and the kind of data we want to handle.

we selected ANNs because they can handle structured data effectively, and our dataset consists of numerical and categorical data. To achieve this task, we use hybrid methodologies, ANN is used for selecting optimized features from patient's data then that optimized features are used to categorize heart disease using the decision tree algorithm. Here we mentioned how a single neuron looks like, what are the parts include that.

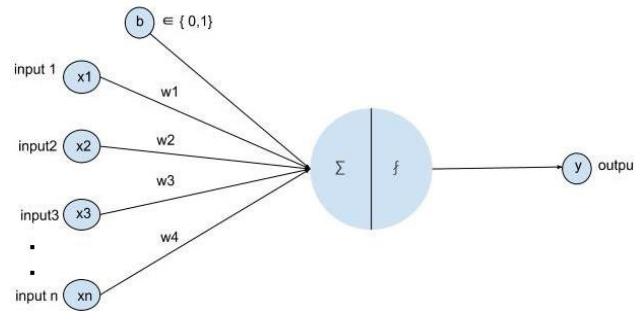


Figure 1: representation of a neuron in a network.

A neuron can process either a single or multiple inputs. In input layer a neuron takes as a single input feature and processed it through transfer function, which calculate the weighted sum of the input along with a bias. The result is then passed into an activation function. The output of each neuron passed to the input of next layers allowing the network to the subsequent layers to process and combine features. Each input is linked to a weight, which determines the its importance and is learned during training, Weights are initialized randomly and adjusted through backpropagation iteratively using the Adam optimizer enables adaptive learning rates for each weight through a stochastic gradient descent approach, combining momentum and RMSProp for faster convergence and stability. In this analysis, the ANN uses Adam optimizer is configured with a learning rate (η) of 0.001 and Binary Crossentropy as the loss function to perform binary classification tasks.

The learning rate controls the step size for weight updates, ensuring efficient optimization. Activation functions like ReLU introduce nonlinearity, Allowing the network to learn intricate patterns and propagate information effectively across layers, forming the foundation of neural computations. The mathematical representation of a neuron with weighted sum, bias and activation function as shown below at each layer in network.

$$y = f\left(\sum_{i=1}^n w_i x_i + b\right)$$

Where,

x_i = inputs features passed to the neuron.
 w_i = Weights are applied to the input.
 b = bias term integrated into the weighted sum.
 $f(.)$ = activation function.
 y = output of the neuron.

As we mentioned earlier to address this problem we use hybrid approach, By using ANN to extract optimized features, those features would be used for classification of heart disease. In this study we used decision tree approach for classifying the heart failure because it gives higher accuracies than other traditional techniques. It classifies the heart failure based upon the features which can be generated by the ANN.

Many research papers conclude that decision tree gives better results than any other traditional techniques but decision tree is unable to handle with noisy data, leads to misclassification or overfitting, to overcome these challenges the input given to the decision tree should be noise free data, so in this case an ANN is helpful to capture the noise free data from the input due to their ability to learn complex patterns. ANN acts as a preprocessing step to improve the performance of decision tree.

4. Feature selection

We used an ANN designed to process and optimize the features from the dataset which has 14 attributes. The ANN has multiple layers to recognize the most relevant patterns. The weights are adjusted during the training of the ANN which helps to reduce the error in predicting heart failure. Higher weights are assigned to features that are mostly contributed to the model performance.

The penultimate layer of the ANN is used as a feature extractor. This layer simplifies the high-dimensional input data into an optimized set of features. Those features help to predict the outcome of heart failure. Based upon certain parameters such as learning rate, step size, and optimizer, the features' patterns may change, which can negatively impact on the model's efficiency. The dimensionality reduction is done by the penultimate layer, not only improving the computational efficiency but also minimizing the risk of overfitting.

This layer compresses the 14 input features into a smaller, optimized set of 8 patterns. These 8 patterns are a combination of several input features. Each pattern or feature represents the higher-level learned relationships and interactions that are more meaningful for predicting the target variable. This dimensionality improvement enhances the computational efficiency by reducing the feature space from 14 to 8, ensuring the model uses only the most useful features while eliminating redundancy and noise. Once the ANN has generated 8 optimized features (patterns), they are passed to the decision tree classifier for prediction.

```
Feature_1 (slope, thal, age)
Feature_2 (thalach, chol, trestbps)
Feature_3 (slope, thal, oldpeak)
Feature_4 (thal, trestbps, chol)
Feature_5 (oldpeak, age, thal)
Feature_6 (cp, restecg, thal)
Feature_7 (thalach, exang, thal)
Feature_8 (ca, trestbps, thalach)
```

Figure 1: Here we mention the 8 patterns or features, which are combinations of the 14 input features from the dataset, based on the weight updates. The combination of several features formed a single pattern, for instance Feature_1 pattern is a combination of slope, thal and age features.

5. Model Evaluation

The assessment of the suggested hybrid model is performed by analysing the performance of the Artificial Neural Network (ANN) and the Decision Tree (DT) classifier under varying conditions. The primary goal is to assess how well the ANN extracts optimized features and how effectively the DT classifies based on these extracted features.

5.1. Performance of ANN

The proposed model employs a type of feedforward neural network, specifically referred to as a Multi-Layer Perceptron (MLP), for feature extraction. The feedforward architecture ensures that data flow in

one direction, from the input layer to output layer through hidden layer.

5.2. Network Architecture

- ✓ **Network Type:** Multi-Layer Perceptron.
- ✓ **Total Layers:** 3.
- ✓ **Input Layer:** Receives 14 attributes from dataset as input features.
- ✓ **Hidden Layer:** Includes neurons with ReLU activation function to introduce nonlinearity and learn complex patterns.
- ✓ **Output Layer:** Extracts 8 optimized features from the penultimate layer.

The network architecture uses the ReLU (Rectified Linear Unit) activation function for the hidden layers, as it has been shown to deliver the maximum average accuracy in comparison to other activation functions. This conclusion aligns with the findings from the base paper, where the authors demonstrated that ReLU performs exceptionally well due to its ability to efficiently handle nonlinearity and avoid the vanishing gradient problem. Based on these insights, ReLU was chosen as the activation function across all hidden layers in this study.

5.3. Testing and Overall Accuracy of ANN

After training our ANN, we tested it on a separate data set to see how well it was working. And we utilized 80% of the data for train the model, while 20% was set aside for testing. The results indicated that our ANN was very accurate, by getting the testing accuracy of 98.54%. Where this means its effectiveness in learning and identifying important patterns in the data. So overall, our ANN was able to predict heart disease outcomes with an accuracy of 99.71%. This determines that our model is robust, efficient, and effective in making predictions.

5.4. Decision Tree Performance

After selecting the optimized features using the ANN, we used the Decision Tree classifier to predict heart disease cases. We tested this model using a separate dataset, and we got the best results. The Decision Tree classifier was more accurate in identifying the people with or without heart disease, with very few mistakes. The results determined high precision and recall, with an overall accuracy of 98.54%. This confirms that the Decision Tree classifier works well with the optimized features selected by the ANN to make the best predictions.

	Precision	Recall	F1-score	Support
0	0.97	1.00	0.99	102
1	1.00	0.97	0.99	103
Accuracy			0.99	205
Macro Avg	0.99	0.99	0.99	205
Weighted avg	0.99	0.99	0.99	205

Table 2 : Here we mentioned the detail classification report of decision tree with their performance.

To assess the performance of my model, I conducted a detailed analysis of its predictions on the test set. The results show a high degree of accuracy, with most predictions being correctly classified. This suggests that the feature extraction and selection process I followed has contributed effectively to the model performance. The detailed breakdown of accuracy, along with other relevant metrics, provides a clear view of how well the model has generalized to the test data. Further details can be seen in the image below.

5.5. Detailed Accuracy Breakdown:

Total Test Samples: 205

Correct Predictions: 202

Incorrect Predictions: 3

Decision tree Accuracy : 98.54%

The Detailed Accuracy Breakdown signifies that the decision tree predicts the correct outcome ,when tested on 20% of dataset, After completing the testing on 205 samples , the decision tree model correctly predicts 202 rows and falsely predicts 3 sample out of 205 sample. It means that if the entire dataset tested

which consists of 1025 rows the decision tree classifier correctly predicts 1010 samples out of 1025 samples.

5.6. Key Findings

The hybrid ANN-DT model achieved remarkable results : The ANN effectively extracted 8 optimized features or patterns from the original dataset, achieving a testing accuracy of 98.54% and an overall accuracy of 99.71%.

The Decision Tree classifier successfully utilized these features, achieving a testing accuracy of 98.54 % and exhibiting high precision and recall.

VI. Discussion of Results with Existing Systems

The results acquired from the hybrid proposed model (ANN+DT) are compared with existing system to evaluate the strength and advantages of using the proposed model. This comparison highlights the improvements achieved from the proposed approach, as well as key aspects such as accuracy, feature optimization and computational efficiency are analyzed to demonstrate how the proposed approach address the limitation of traditional methods and increase predictive performance.

Table 3 : comparing the accuracy of hybrid proposed model with existing system accuracies.

Model	Accuracy
Naïve Bayes	79%
Logistic Regression	81%
K-Nearest Neighbor	86%
SVM	89%
AdaBoost	89%
Traditional Decision tree	85-95%
ANN	86%
CPT+	90.5%
K-Means	76%
Proposed Hybrid Model (ANN+DT)	98.54%

The results showned in table clearly highlight the superior performance of proposed hybrid system than existing results for heart disease prediction with an accuracy of 98.54%. The hybrid model significantly performs better than traditional techniques including decision tree, svm, k-means and cpt+ achieves accuracies ranging from 85% to 92% this improvement can only obtained through Ann. Its ability to extract optimized and relevant features from the dataset effectively and minimize the redundancy. The merging of feature extraction and classification into a hybrid approach ensures both efficiency and accuracy. these finding highlight the practical implication of the proposed system. on more important advantage is taking less computational cost than ensemble methodologies which was used in existing system however, it is importance to acknowledge that different variations in dataset size or preprocessing techniques used by existing methods may have influenced by their results and these factors should be considered when interpreting the comparison.

VII. Visualization

The feature importance graph illustrates the contribution of each extracted feature in the Decision Tree model for heart failure prediction. The graph effectively highlights the hierarchical importance of features, showcasing how specific combinations of clinical parameters enhance the model's decision-making capability during the feature extraction process.

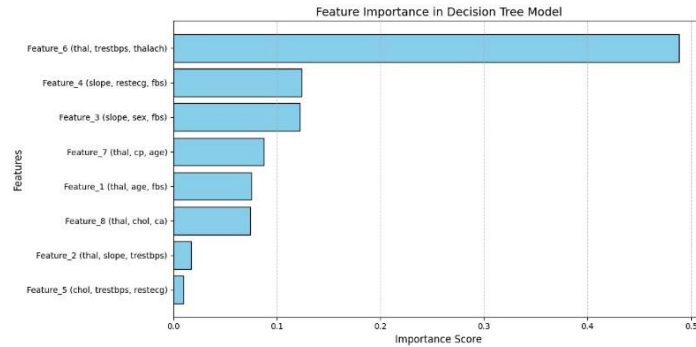


Figure 2: In the above figure shows the Feature_6(thal, trestbps,thalach) plays a major role in decision tree to classify the heart failure,Feature_4(slope,restecg,fbs) and Feature_3(slope,s ex,fbs) plays a second-highest importance followed by Feature_7(thal,cp,age),Feature_1(thal,age,fbs),Feature_6(thal,chol,ca),Feature_2(thal,slope,trestbps)and Feature_5(chol,trestbps,restecg) which plays a minor roles for predicting the heart failure in view of decision tree.

The correlation heatmap helps us to find out how different features are related to each other. By observing it, we can identify features which are very similar and which can negatively impact on our model's performance. By removing unnecessary features, we can keep only the most important ones which provides the most value. This helps us to simplify our data and improve our model's speed and accuracy By removing unnecessary features, we can keep only the most important ones which provides the most value. This helps us to simplify our data and improve our model's speed and accuracy.

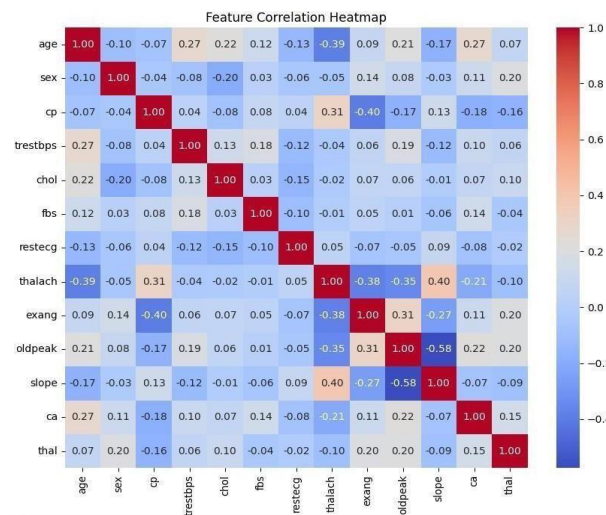


Figure 6: correlation heatmap shows the relationship between the features.

Conclusion

In This hybrid approach, Integration of ANN for the best feature extractor and a decision tree classifier for heart disease prediction, provides a better and realistic solution for early detection. The ANN systematically bring out the noise free and sequent features, and enhance the decision tree to provide accurate and interpretable predictions. This structured integration improves the systematic efficiency whenever addressing challenges in traditional methods. The future work could explore on integrating real time monitoring data and electronic health records to improve the model's robustness. Testing on larger dataset across various population groups can increase generalizability.

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